

AKSHAYA BHARADHWAJ

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EDUCATION

University of California, Irvine, CA

Master of Data Science, GPA: 3.90/4.00

December 2026

Relevant Coursework: Intermediate Probability and Statistics I, Introduction to AI, Databases and Data Management

Mukesh Patel School Of Technology, Management & Engineering, SVKMS's NMIMS, Mumbai, India

Bachelor of Technology, Data Science, GPA: 3.74/4.00

August 2025

Relevant Coursework: Statistics, Data Visualization and Analysis, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

TECHNICAL SKILLS

Languages: Python, SQL, R, HTML/CSS, JavaScript

Machine Learning / GenAI: NLP, LLMs (multi-agent), deep learning, model evaluation, error analysis, A/B testing

Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, Librosa, Matplotlib, Seaborn, Flask, Streamlit

MLOps / Dev: Git

Data: SQL, MySQL

Tools: LM Studio, Tableau, PowerBI, MySQL, Google Colab, Microsoft Excel, Postman

PROFESSIONAL EXPERIENCE

WIPRO LIMITED

Research Intern - Machine learning at Wipro Infrastructure Engineering (Wipro PARI) - Remote

December 2024 – August 2025

- Built and evaluated ANN models using Python and scikit-learn, validated using R^2 , to model complex Li-ion battery manufacturing processes, enabling analysis of how upstream parameters impact downstream material quality and reliability.
- Developed a process-parameter framework spanning 16 manufacturing stages in collaboration with industry experts, supporting data-driven process optimization across the EV battery pipeline.
- Contributed to 2 review papers and a research paper proposing a machine-learning-driven framework for EV battery process optimization, currently under submission to academic and industry venues.

Data Science Intern - Wipro Water, Navi Mumbai, India - Onsite

May 2024 – July 2024

- Analyzed and cleaned datasets from urban commercial projects across multiple major Indian cities, uncovering trends to support business decision-making.
- Translated analytical insights into business inputs by collaborating with the marketing team and senior leadership to support sales and strategy decisions.
- Built Python-based automation scripts to clean, validate, and format raw data for Salesforce lead creation, automating a previously manual process and improving operational efficiency.

Intern, IDEA CREST SOLUTIONS PRIVATE LTD, Bangalore, India - Remote

May 2023 – July 2023

- Developed RESTful APIs using Python (Flask) and optimized MySQL database schemas for a streaming platform management system, improving data integrity and query performance.
- Created Postman collections to test endpoints and document API workflows, improving handoffs and reducing debugging time for integrations.

RESEARCH & PROJECT EXPERIENCE

Skin Cancer Detection from Dermoscopic Images | ML Course Project | COMPSCI 273P

March 2026

- Developed a multimodal CNN + MLP pipeline on 400k+ dermoscopic images (ISIC 2024), achieving 0.93 recall and 0.942 ROC-AUC under 0.1% class imbalance.

Raga Recognition from Audio: End-to-End ML Classification | DATA 210P Statistical Methods I Course Project

March 2026

- Engineered a 228-dimensional hybrid feature vector and trained 5 classifiers (SVM, RF, XGBoost, MLP, Ensemble) on the Saraga Carnatic dataset, achieving 77.8% accuracy (XGBoost) vs 33.3% random baseline under severe data scarcity using GroupKFold cross-validation.

ValidClean: Multi-Agent Agentic AI Framework for Data Preprocessing- Presented at UCI GradSLAM 2026

February 2026 – Present

- Architected a 5-agent LLM system to autonomously profile datasets, infer ML objectives, and generate preprocessing strategies, validated using K-fold cross-validation, paired t-tests, and Cohen's d.

Brain MRI Classification & Segmentation | Advanced Computer Vision Project

November 2024

- Implemented a U-Net architecture (4 contracting + 4 expanding blocks) for brain tumor segmentation from MRI scans, achieving 80% test accuracy in delineating tumor regions from paired ground truth masks.
- Fine-tuned YOLOv8 for brain tumor detection and classification, outperforming VGG-16 baseline; explored autoencoder-based anomaly detection for reconstruction-based tumor identification.

Automated Web Data Extraction via Zero-Shot NLP Classification | [Published](#) | ICDAM-2025

Nov 2024 – May 2025

- Built a data extraction system using Zero-Shot Learning and NLP classification; published in Elsevier SSRN and presented at ICDAM-2025.